

VECTOR OPERATIONS (**)

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A vector operator is a transformation of vectors into vectors: $A: \mathcal{E}^3 \rightarrow \mathcal{E}^3$ (*)

DEF $A: \mathcal{E}^3 \rightarrow \mathcal{E}^3$ is a vector operator

A is linear if $A(\alpha \underline{u} + \beta \underline{v}) = \alpha A(\underline{u}) + \beta A(\underline{v})$

A is symmetric if $\underline{u} \cdot A(\underline{v}) = \underline{v} \cdot A(\underline{u})$

A is anti-symmetric if $\underline{u} \cdot A(\underline{v}) = -\underline{v} \cdot A(\underline{u})$

Properties

- If A is symmetric (or anti-symmetric) then it is linear

sym/anti-sym

Proof $\underline{u} \cdot A(\alpha \underline{u} + \beta \underline{w}) = \pm (\alpha \underline{v} + \beta \underline{w}) \cdot A(\underline{u}) =$
 $= \pm \alpha \underline{v} \cdot A(\underline{u}) \pm \beta \underline{w} \cdot A(\underline{u}) =$
 $= \underline{u} \cdot [\alpha A(\underline{v}) + \beta A(\underline{w})] \quad \forall \underline{u}, \underline{v}, \underline{w}$

(*) $\underline{a} \times \underline{b}$ can be seen as $A(\underline{b})$
 $\text{rot}_{\underline{a}}(\underline{b})$ can be seen as $R_{\underline{a}}(\underline{b})$ rotation

(**) Functions of the kind $\underline{v}: \mathcal{E}^3 \rightarrow \mathcal{E}^3$

- If $\underline{u} \cdot A \underline{u} = 0 \quad \forall \underline{u} \in \mathbb{E}^3$ then A is anti-symmetric 11/

Proof Consider $0 = (\underline{u} + \underline{v}) \cdot A(\underline{u} + \underline{v}) =$
 $= (\underline{u} + \underline{v}) \cdot [A(\underline{u}) + A(\underline{v})] =$ ↑ linearity
 $= \underline{u} \cdot A(\underline{u}) + \underline{v} \cdot A(\underline{u}) + \underline{u} \cdot A(\underline{v}) + \underline{v} \cdot A(\underline{v})$
0 0

Then: $\underline{v} \cdot A(\underline{u}) + \underline{u} \cdot A(\underline{v}) = 0 \Leftrightarrow \underline{v} \cdot A(\underline{u}) = -\underline{u} \cdot A(\underline{v})$
↑
 A is anti-symmetric

- If A is anti-symmetric $\exists! \underline{a} \in \mathbb{E}^3$ s.t.
 $A(\underline{u}) = \underline{a} \times \underline{u} \quad \forall \underline{u} \in \mathbb{E}^3$

Proof Consider a basis $\{\underline{e}_1, \underline{e}_2, \underline{e}_3\}$ and define

$$\underline{a} \triangleq \frac{1}{2} \sum_k \underline{e}_k \times A(\underline{e}_k)$$

Compute $\underline{a} \times \underline{u} = \frac{1}{2} \sum_k [\underline{e}_k \times A(\underline{e}_k)] \times \underline{u}$ double
non
product

$$= \frac{1}{2} \sum_k \left[\underbrace{(\underline{u} \cdot \underline{e}_k)}_{\underline{u}_k} A(\underline{e}_k) - \underbrace{(\underline{u} \cdot A(\underline{e}_k))}_{-\underline{e}_k \cdot A(\underline{u})} \underline{e}_k \right] =$$

↑
 A is antisymmetric

$$= \underbrace{\frac{1}{2} \sum_k m_k A(\underline{e}_k)}_{\substack{\uparrow \text{linearity} \\ \frac{1}{2} \sum_k A(m_k \underline{e}_k) \\ \uparrow \text{linearity} \\ \frac{1}{2} A(\underline{u})}} + \underbrace{\frac{1}{2} \sum_k [A(\underline{u}) \cdot \underline{e}_k] \underline{e}_k}_{\substack{\uparrow \\ \frac{1}{2} A(\underline{u})}} =$$

$$= \frac{1}{2} A(\underline{u}) + \frac{1}{2} A(\underline{u}) = A(\underline{u})$$

EXERCISE
Proof uniqueness
of a

ALGEBRAIC FORM OF OPERATORS

Linear operators can be represented as matrices when expressed in algebraic form

For a generic vector $\underline{u}$ and a basis $\{\underline{e}_1, \underline{e}_2, \underline{e}_3\} = \langle \mathcal{O} \rangle$

$$\begin{aligned} \underline{v} = A(\underline{u}) &= \sum_i [\underline{e}_i \cdot A(\underline{u})] \underline{e}_i = \sum_{ij} [\underbrace{\underline{e}_i \cdot A(\underline{e}_j)}_{\substack{\uparrow \\ \sum_j m'_j \underline{e}_j}} m'_j] \underline{e}_i = \\ &= \sum_i \left(\sum_j A_{ij} m'_j \right) \underline{e}_i \\ &\quad \uparrow \\ \sigma_i &\triangleq (A \cdot \underline{m})_i \quad \text{or } i\text{-th component} \end{aligned}$$

double sums

NOTE

$$\underline{v} = A(\underline{u}) \iff {}^0\underline{v} = {}^0A {}^0\underline{u}$$

↑

$${}^0A = \begin{bmatrix} \underline{e}_1 \cdot A(\underline{e}_1) & \underline{e}_1 \cdot A(\underline{e}_2) & \underline{e}_1 \cdot A(\underline{e}_3) \\ \underline{e}_2 \cdot A(\underline{e}_1) & \underline{e}_2 \cdot A(\underline{e}_2) & \underline{e}_2 \cdot A(\underline{e}_3) \\ \underline{e}_3 \cdot A(\underline{e}_1) & \underline{e}_3 \cdot A(\underline{e}_2) & \underline{e}_3 \cdot A(\underline{e}_3) \end{bmatrix}$$

If A is symmetric then ${}^0A = {}^0A^T$

If A is anti-symmetric then ${}^0A = -{}^0A^T$ $\leftarrow A^0$ is skew-symmetric (*)

Proof A symmetric then

$$[A]_{ij} = \underline{e}_i \cdot A(\underline{e}_j) = \underline{e}_j \cdot A(\underline{e}_i) = [A]_{ji}$$

A anti-symmetric then

$$[A]_{ij} = \underline{e}_i \cdot A(\underline{e}_j) = -\underline{e}_j \cdot A(\underline{e}_i) = -[A]_{ji}$$

(*) If $M = -M^T$ then $M = \begin{bmatrix} 0 & -a & -b \\ a & 0 & -c \\ b & c & 0 \end{bmatrix}$ and

$$M^T = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix} = -M$$

COORDINATE TRANSFORMATIONS FOR OPERATORS

Consider $\langle a \rangle = \{\underline{e}_1^a, \underline{e}_2^a, \underline{e}_3^a\}$ and $\langle b \rangle = \{\underline{e}_1^b, \underline{e}_2^b, \underline{e}_3^b\}$

$$\underline{v} = A(\underline{u}) \iff {}^a \underline{v} = {}^a A {}^a \underline{u}$$

$$\underline{v} = A(\underline{u}) \iff {}^b \underline{v} = {}^b A {}^b \underline{u} \quad \odot$$

$${}^a \underline{v} = {}^a {}^b R {}^b \underline{v} \quad \otimes$$

$${}^a \underline{u} = {}^a {}^b R {}^b \underline{u} \iff {}^b \underline{u} = ({}^a {}^b R)^{-1} {}^a \underline{u} = {}^a {}^b R^T {}^a \underline{u} = {}^b {}^a R {}^a \underline{u} \quad \Delta$$

Then :

$${}^a \underline{v} = {}^a {}^b R {}^b \underline{u} = {}^a {}^b R {}^b A {}^b \underline{u} = ({}^a {}^b R {}^b A {}^b {}^a R) {}^a \underline{u} \quad \otimes \quad \Delta$$

Therefore, the operator matrix ${}^b A$ can be transformed into operator matrix ${}^a A$ as

$${}^a A \triangleq {}^a {}^b R {}^b A {}^b {}^a R$$

or vice versa

$${}^b A = ({}^a {}^b R)^T {}^a A ({}^b {}^a R)^T = {}^b {}^a R {}^a A {}^a {}^b R$$

NOTE Vector "notations" are transformations "from $\rightarrow$ to"
Operator "notations" are transformations "from $\rightarrow$ to $\rightarrow$ from"

ADDEDUM (more on vector operations)

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ALGEBRAIC FORM OF CROSS PRODUCT

Consider $(\underline{a} \times \underline{b})$ for any $\underline{a}, \underline{b}$

$$\begin{aligned}(\underline{a} \times \underline{b}) &= [a_1 \underline{e}_1 + a_2 \underline{e}_2 + a_3 \underline{e}_3] \times [b_1 \underline{e}_1 + b_2 \underline{e}_2 + b_3 \underline{e}_3] \\&= [a_1 (\underbrace{\underline{e}_1 \times \underline{e}_1}_0) + a_2 (\underbrace{\underline{e}_2 \times \underline{e}_1}_{-\underline{e}_3}) + a_3 (\underbrace{\underline{e}_3 \times \underline{e}_1}_{\underline{e}_2})] b_1 + \\&\quad + [a_1 (\underbrace{\underline{e}_1 \times \underline{e}_2}_{\underline{e}_3}) + a_2 (\underbrace{\underline{e}_2 \times \underline{e}_2}_0) + a_3 (\underbrace{\underline{e}_3 \times \underline{e}_2}_{-\underline{e}_1})] b_2 + \\&\quad + [a_1 (\underbrace{\underline{e}_1 \times \underline{e}_3}_{-\underline{e}_2}) + a_2 (\underbrace{\underline{e}_2 \times \underline{e}_3}_{\underline{e}_1}) + a_3 (\underbrace{\underline{e}_3 \times \underline{e}_3}_0)] b_3 \\&= (a_2 b_3 - a_3 b_2) \underline{e}_1 + (a_3 b_1 - a_1 b_3) \underline{e}_2 + (a_1 b_2 - a_2 b_1) \underline{e}_3\end{aligned}$$

Then considering the ^{coordinate} frame $\langle \underline{e} \rangle = \{ \underline{e}_1, \underline{e}_2, \underline{e}_3 \}$

$${}^o(\underline{a} \times \underline{b}) = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$